Modelling a 3D printer

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October 14, 2026

0 Introduction

3D Printers have the basic task of extruding filament at a given x, y, z coordinate at time t. The mechanism that achieves that movement determines the machine's costs, print quality, and robustness. Out of many different options, the Core XY mechanism is a popular choice for modern printers. Different than Cartesian designs, where one motor drives one axis, here each motor simultaneously changes the x and y position of the tool head (see Figure 1). This results in a more complex kinematic model, but allows a more sturdy casing and faster printing speeds. Our task is to model and simulate the system.

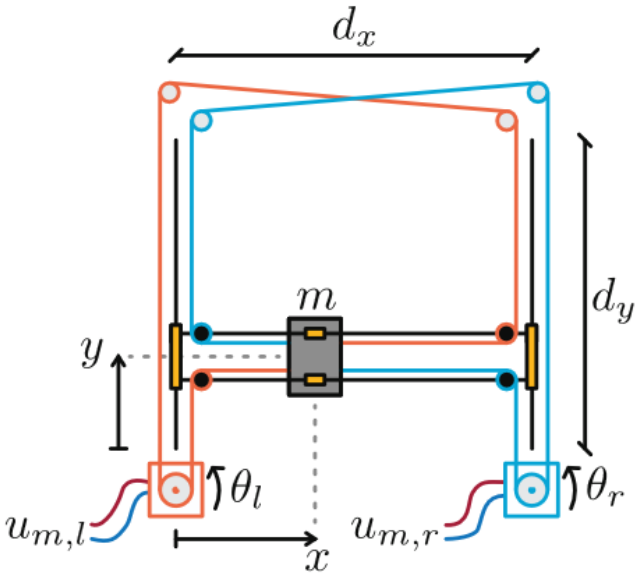


Figure 1: Schematic of the Core XY mechanism.

0.1 Symbol table

Table 1: Symbols and parameter values.

Description	Symbol	Value
Motor angle left	θ_l	–
Motor voltage left	$u_{m,l}$	–
Motor current left	$i_{m,l}$	–
Motor radius	r_m	$11\,\mathrm{mm}$
Motor induction	L_m	$10\,\mathrm{mH}$
Motor resistor	R_m	$90\,\Omega$
Motor inertia	J_m	$8 \times 10^{-6}\,\mathrm{kg\,m^2}$
Motor constant	K_m	$0.07\,\mathrm{N\,m/A}$
Tool head position	x, y	–
Tool head mass	m	$60\,\mathrm{g}$
Nozzle voltage	u_n	–
Nozzle current	i_n	–
Nozzle capacitor	C_n	–
Nozzle thermal capacity	$C_{t,n}$	–
Nozzle thermal resistance	$R_{t,n}$	$9\,\mathrm{K/W}$
Nozzle resistor	R_n	$8\,\Omega$
Nozzle temperature	T_n	$240\,^\circ\mathrm{C}$
Environmental temperature	T_e	$300\,\mathrm{K}$
Bearing friction rolling	μ_r	$0.003\,\mathrm{N\,s/m}$
Bearing friction linear x-direction	$\mu_{l,x}$	$0.002\,\mathrm{N\,s/m}$
Bearing friction linear y-direction	$\mu_{l,y}$	$0.001\,\mathrm{N\,s/m}$
Bearing radius	r_b	$3\,\mathrm{mm}$
Print bed temperature	T_b	$105\,^\circ\mathrm{C}$
Print area dimension x	d_x	$29\,\mathrm{cm}$
Print area dimension y	d_y	$29\,\mathrm{cm}$

1 Physical Modelling

1.1 Kinematics

2 Finite State Machine

3 Hybrid System

References